Task 1: Five-minute demo video or live walkthrough

Time	Demonstration	What to show
0:00–0:45	Introduction	Explain your secure cloud architecture and the resources you created.
0:45–1:45	Access security	Show an IAM policy or EC2 security group that restricts access.
1:45–2:45	Data protection	Show S3 server-side encryption and the bucket's access settings.
2:45–4:00	Monitoring	Open CloudWatch and show the EC2 CPU utilization graph and available logs.
4:00–5:00	Alerting and conclusion	Show your CPU alarm, its threshold, and the configured email notification. Explain the security benefits.

Task 2: Written peer feedback

Peer Review: Secure Cloud Infrastructure

1. One thing done well in security:
The architecture uses restricted access permissions and network security rules to reduce unauthorized access to cloud resources.

2. One suggestion for improving monitoring:
Add CloudWatch alarms for important events, such as high CPU utilization, repeated application errors and suspicious access activity. Configure notifications so that the administrator can respond promptly.

3. Questions about the implementation:
- Are the application servers isolated from direct public internet access?
- Are logs retained securely and reviewed regularly?

- Has the email notification system been tested to confirm that alerts are received?

Task 3: Update the infrastructure based on feedback

Item	Before change	After change	Evidence
Security group	An unnecessary inbound rule may be present	Remove it and allow only required traffic	Before/after security group screenshots
CloudWatch monitoring	Only basic metrics are available	Enable additional metrics or application logs as needed	CloudWatch metrics and log group screenshots
CPU alert	No threshold alarm is configured	Configure an alarm for CPU utilization above 70%	Alarm configuration screenshot
Notifications	Email notification has not been tested	Confirm subscription and test notification delivery	SNS subscription and test evidence

Task 4: Reflection (5–6 sentences)

One challenge I faced while securing my cloud infrastructure was understanding how to configure IAM permissions without giving unnecessary access to users. I addressed this by reviewing the required actions and restricting permissions to the specific resources needed for the task. I also learned that monitoring CPU utilization alone is not enough to detect every security incident. CloudWatch logs and alarms can improve visibility and help identify unusual activity. In the future, I would test my security policies and notification settings before deploying the application. I

would also review access permissions regularly and document all configuration changes.